

SOFTWARE REQUIREMENTS SPECIFICATION

Smart Electricity Consumption Prediction System

Faculty of Computers and Information
Damanhour University, Egypt

Document	Software Requirements Specification (SRS)
Version	1.1 — Updated Stakeholders & Users table
Scope	Selected spaces within the Faculty of Computers and Information
Status	Proposed requirements baseline

Planning note: confirm stakeholder roles, data access, and available electricity measurements with the faculty before implementation.

Table of Contents

- 1. Introduction
- 2. Project Overview
- 3. Problem Statement
- 4. Objectives
- 5. Stakeholders & Users
- 6. Functional Requirements
- 7. Non-Functional Requirements
- 8. Data Requirements
- 9. Project Scope
- 10. System Flow
- 11. Machine Learning Approach
- 12. ML Pipeline
- 13. Requirements Traceability Matrix
- 14. Technology Stack
- 15. Assumptions & Constraints
- 16. Requirements Phase Deliverable

The final ML model and classification thresholds will be chosen from data analysis and evaluation, not assumed in advance.

1. Introduction

This SRS describes the proposed requirements for a Smart Electricity Consumption Prediction System. It is intended to help users analyze electricity-related data and estimate consumption for selected spaces within the Faculty of Computers and Information at Damanhour University.

The document establishes a shared understanding before detailed analysis, design, implementation, and testing. It may be refined after stakeholder consultation and data assessment.

2. Project Overview

The proposed web application accepts information about a selected location and its operating conditions, applies a trained machine-learning model, and displays estimated electricity consumption in kWh. It may also classify consumption, show charts and historical information, and provide energy-saving recommendations. Candidate spaces include computer laboratories, lecture halls, and offices; actual locations depend on data access and faculty confirmation.

3. Problem Statement

There is a need to better understand electricity-consumption patterns in selected faculty spaces and estimate expected consumption using available data. Organized analysis and prediction can help compare patterns across locations and support energy-conscious decisions. The proposed system offers data-driven estimates; it does not control devices or guarantee energy savings.

4. Objectives

- Collect or organize suitable consumption and operating-condition data for selected spaces.
- Clean and prepare the dataset for analysis and machine learning.
- Explore patterns and relationships in available data.
- Train and compare regression models for consumption prediction.
- Evaluate models using MAE, RMSE, and R^2 where appropriate.
- Display predictions and summaries in a user-friendly dashboard.
- Provide evidence-based recommendations where appropriate.
- Document requirements, design, implementation, and testing.

5. Stakeholders & Users

Stakeholders have an interest in, or may be affected by, the system. Users interact with it directly. A user can also be a stakeholder. The roles below are proposed and should be confirmed with the faculty.

Stakeholder / User	Relationship	Role / Interest
Faculty staff / authorized user	Direct user	Select location, enter/review data, request predictions, view results.
Faculty administration	Stakeholder; possible report user	Use summaries to understand patterns and support planning decisions.
Maintenance / facilities personnel	Stakeholder; possible data provider/user	Help identify available meter readings and interpret patterns.
Project team	Developer / maintainer	Build, test, document, and maintain the system.
Faculty representative / supervisor	Stakeholder / reviewer	Confirm scope, data access, and whether proposed features meet faculty needs.

Not every stakeholder needs to log in. Access permissions will be specified if authentication or role-based access is included.

6. Functional Requirements

Functional requirements describe actions the system must perform.

ID	Requirement	Description
FR-01	Data Input	Allow the user to enter required operating-condition data for a prediction.

ID	Requirement	Description
FR-02	Location Selection	Allow selection of a supported faculty location.
FR-03	Input Validation	Check required fields, data types, and reasonable ranges; show clear error messages.
FR-04	Data Preprocessing	Apply preprocessing required by the trained model, consistently with training.
FR-05	Consumption Prediction	Use the selected regression model to estimate consumption in kWh.
FR-06	Display Prediction	Present predicted consumption clearly, including the unit.
FR-07	Consumption Classification	Optionally classify as Low, Medium, or High using thresholds justified by data.
FR-08	Dashboard	Show charts and summaries for available predictions or records.
FR-09	Recommendations	Display practical recommendations tied to observed patterns; avoid unsupported savings claims.
FR-10	Model Evaluation	Compare candidate models using suitable metrics during development.
FR-11	New Prediction	Allow another prediction without restarting the application.
FR-12	Historical Data	If storage/data access are implemented, allow viewing prior records or predictions.

FR-10 is primarily a development activity; whether metrics are displayed to end users will be decided during design.

7. Non-Functional Requirements

ID	Attribute	Requirement
NFR-01	Usability	Use clear labels, instructions, and readable results.
NFR-02	Performance	Return predictions within a reasonable response time in the target environment.
NFR-03	Reliability	Handle invalid input and predictable failures gracefully.
NFR-04	Maintainability	Keep code modular, readable, and organized for easier updates.
NFR-05	Scalability	Allow additional supported locations or records without major redesign.
NFR-06	Prediction Quality	Evaluate held-out performance using MAE, RMSE, and R^2 where appropriate; do not claim a target before testing.
NFR-07	Security	If data is stored or access restricted, apply appropriate access controls and data protection.
NFR-08	Browser Compatibility	Test on commonly used modern browsers supported by the deployment environment.

8. Data Requirements

The dataset should contain relevant features and a reliable target. Confirm the availability of each field before model development.

Field	Purpose
Date	Date/time associated with the record; supports temporal analysis.
Location	Identifies the measured space.
Temperature / Humidity	Environmental features if reliably available.
Occupancy	Number of people or a consistent occupancy measure.
Number_of_Computers	Count of computers or relevant devices.
Operating_Hours	Duration the location is operating.
AC_Hours / Lighting_Hours	Operating durations if available.
Previous_Consumption	Prior measured consumption if available and correctly aligned.
Electricity_Consumption	Target variable: measured electricity consumption in kWh.

Data quality: check missing values, duplicates, inconsistent units, and implausible values. Investigate outliers rather than automatically deleting them. Use actual meter readings where possible. If per-room readings are unavailable, state the measurement level clearly and never present invented room-level values as real.

The proposed collection period and number of locations are planning targets, not confirmed facts.

9. Project Scope

In Scope

- Data collection/organization for selected spaces, subject to access.
- Data cleaning, exploratory analysis, and preprocessing.
- Training and comparing regression models.
- Prediction and optional consumption classification.
- Web interface, dashboard, visualizations, recommendations, and historical analysis where records exist.
- Software testing and documentation.

Out of Scope

- Automatic control or shutdown of air conditioners, computers, or lights.
- Full smart-grid or university electricity-network control.
- Electricity billing or payment functionality.
- Real-time IoT control or sensor installation as a required feature.
- Management of every university building; scope is selected spaces within the Faculty of Computers and Information.

10. System Flow

High-level flow:

```
User opens web interface
↓
Selects a supported location
↓
Enters or selects required data
↓
System validates input
↓
Preprocessing compatible with trained model
↓
ML model estimates consumption
↓
System displays predicted kWh
↓
Optional classification
↓
Dashboard and recommendations
```

This is a high-level SRS flow. Detailed interactions, branches, and error paths will be modeled during System Analysis.

11. Machine Learning Approach

The primary task is regression because the output is a numeric consumption estimate in kWh. Candidate models:

- Linear Regression: a simple baseline for linear relationships.
- Decision Tree Regressor: rule-based splits that can model non-linear relationships.
- Random Forest Regressor: combines multiple decision trees and may improve robustness.

Compare models using an appropriate data split and metrics such as MAE, RMSE, and R^2 . Choose the final model based on evaluation, data quality, and practical considerations.

12. ML Pipeline

```
Data collection → data quality checks and cleaning → missing-value/outlier investigation → feature
selection/engineering → train/test split → preprocessing and model training → evaluation on held-out
data → model comparison/selection → save and use selected model for prediction.
```

Preprocessing steps fitted from data (such as imputation or scaling) must be fitted on training data only and then applied to test/new data to avoid data leakage.

13. Requirements Traceability Matrix

This matrix connects requirements to design, implementation, and testing. Entries are an initial plan; detailed identifiers will be added as artifacts are created.

Requirement	Design Evidence	Implementation Evidence	Test Evidence
FR-01 Data Input	Input form design	Data-entry interface	TC-01: valid/missing input
FR-02 Location	Location selector design	Location selector	TC-02: supported location
FR-03 Validation	Validation rules	Validation logic	TC-03: invalid values
FR-05 Prediction	Prediction component	Model inference function	TC-05: output and unit
FR-06 Result	Result view design	Result display	TC-06: clear result
FR-08 Dashboard	Dashboard layout	Charts/summaries	TC-08: correct chart data
FR-12 History	History/storage design if included	History view if included	TC-12: correct prior records

14. Technology Stack

Technology	Purpose
Python	Primary language for data preparation and ML.
Pandas / NumPy	Tabular data handling and numerical operations.
Scikit-learn	Model training, preprocessing, and evaluation.
Streamlit	Proposed web interface and interactive dashboard.
Figma	Wireframes and UI/UX planning.
Git / GitHub	Version control, collaboration, and source hosting.
CSV or database	Storage choice based on actual needs and history features.

15. Assumptions & Constraints

Assumptions

- Suitable consumption measurements can be obtained for at least some selected spaces.
- Available features may contain useful information for prediction.
- Users enter valid data for new predictions.
- The project environment can run the selected Python libraries.

Constraints

- Actual meter readings and measurement granularity may be limited.
- Data quantity and quality may limit predictive performance.
- Time, team resources, and deployment environment are limited by the academic schedule.
- The model may not generalize to unrepresented locations or conditions.
- The system provides estimates and decision support; it does not guarantee savings or replace professional electrical assessment.

16. Requirements Phase Deliverable

The main deliverable is this SRS, documenting the problem, objectives, stakeholders/users, functional and non-functional requirements, data needs, scope, high-level flow, ML approach, technology proposal, assumptions, and constraints.

After reviewing and agreeing on the requirements, the next phase is System Analysis. Planned artifacts include a Use Case Diagram, detailed Use Case Descriptions, an Activity Diagram, and an initial system-module breakdown. Later phases cover design, implementation, testing, and final documentation.

Review note: confirm stakeholder roles, data access, measurement units, selected locations, and optional features with the faculty before treating this document as final.